

PAPER_959: 26D Ramanujan Summation Engine

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sion: 215 **Source:** ramanujan_26d_summation.py (Ramanujan26DSummation) **Calcu-**
lator: Ramanujan26DSummationCalc (CP4 #543) **CVW:** v2.0.0 compliant

Abstract

We implement the full 26-dimensional Ramanujan-accelerated polylogarithm summation $S_{26}(z) = \sum_{n=1}^N z^n / n^{26} \cdot R_n^{(26)}$. The acceleration factor $R_n^{(26)}$ achieves convergence to machine precision in ≤ 50 terms. At $z = [SSq] = 0.57$, the result matches $\text{Li}_{26}(0.57)$ exactly, driving all E(t), phonon, jet, and NS effects.

1. Ramanujan Acceleration Factor

$$R_n^{(26)} = \frac{1}{n!} \left(1 + \frac{1}{n^{26}} \sum_{k=1}^{26} (-1)^{k+1} \binom{26}{k} \frac{(26-k)!}{n^k} \right)$$

2. 26D Summation

$$S_{26}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}} \cdot R_n^{(26)}$$

3. Convergence

At $z = 0.57$, $N = 50$: converges to full machine precision.

References

1. Ramanujan, S. - Collected Papers (1927)
2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)